👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.1

Determine the Area of Triangles

Lesson 5-2 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

SKY HARBOR MISSION

The Triangular Skylight

You are the lead architect at Sky Harbor Studio. The city has hired your firm to design a glass skylight made of triangular panels for the new transit hub. Before the glass can be cut, you must master the area of triangles — every wrong measurement wastes expensive glass.

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the area of a triangle using the formula A = ½ × base × height.

Language Objective: I can explain my steps using the words base, height, area, and perpendicular.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Base **Height** **Area** **Perpendicular** **Composite figure**

Formula **Parallel** **Substitute**

Level 3: try the blanks from memory first, then check the bank.

1. The side of a triangle you measure the height from at a 90° angle is the _____ .

2. The perpendicular distance from the base to the opposite vertex is the _____ .

3. The amount of flat space inside a triangle is its _____ .

4. Two lines that meet at a 90° angle are _____ .

5. A shape made of two or more basic shapes combined is a _____ .

6. A math rule written with symbols, like A = ½ × b × h, is a _____ .

7. Two sides that stay the same distance apart and never meet are _____ .

8. Putting known numbers into a formula is called _____ .

Write About the Math

How many cans of paint does the homeowner need?
¿Cuántas latas de pintura necesita el dueño de la casa?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Base** — *Base*
- ☐ **Height** — *Altura*
- ☐ **Area** — *Área*
- ☐ **Perpendicular** — *Perpendicular*
- ☐ **Composite figure** — *Figura compuesta*

Model: *The height is the perpendicular distance from the base to the top corner.* — Students identify the height as the perpendicular (straight-up, 90-degree) distance from the base to the opposite vertex, not the slanted side.

Start

Complete the frames. Use the word bank.
Completa las oraciones. Usa el banco de palabras.

- **The wall's area is ___ sq ft because A = 1/2 × ___ × ___. She needs ___ cans because ___ ÷ 50 = ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.

- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Base
2. **Notes 2:** Height
3. **Notes 3:** Area
4. **Notes 4:** Perpendicular
5. **Notes 5:** Composite figure
6. **Notes 6:** Formula
7. **Notes 7:** Parallel
8. **Notes 8:** Substitute
9. **Watch for this mistake:** *A common mistake in Area of Triangles is forgetting the ½ — students multiply base × height like a rectangle and stop there, or they use the triangle's slanted side instead of the perpendicular height that's already given. Always check: did I take half, and did I use the straight-up height, not the slanted side?*
10. **Try It 1:** B. 14 sq ft — *A = ½ × base × height = ½ × 7 × 4 = ½ × 28 = 14 sq ft. Multiply base × height first (28), then take half (14).*
11. **Try It 2:** C. 33 sq ft — *A = ½ × base × height = ½ × 11 × 6 = ½ × 66 = 33 sq ft.*